AIC-607 Brain-Computer Interface

Classification & Interpretation Report

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Part 1: Problem Setup (20 Marks)

Q1: How many epochs are in each class and whether they are balanced

After merging the epochs, there are 47 eyes_open epochs and 50 eyes_closed epochs, resulting in a total of 97 epochs. The eyes_open class combines 23 epochs from eyes_open_1 and 24 from eyes_open_2, and the eyes_closed class combines 28 from eyes_closed_1 and 22 from eyes_closed_2. The dataset is well balanced, with a class ratio of 47:50 (≈0.94), which is close to 1. This means the classifier will see roughly equal amounts of both classes during training, so it cannot just always guess one class and get a good score.

Q2: What the chance level is for this 2-class problem and what it means

For this 2-class classification problem, the chance level is 50%. In our dataset, the two classes are nearly equal, 47 for eyes_open and 50 eyes-closed epochs. So 50% is a fair baseline. Therefore, any model developed for this task must exceed 50% accuracy to demonstrate meaningful learning and the ability to distinguish between the two classes.

Q3: Why you are using only these two classes and not all five blocks

Only the eyes_open and eyes_closed conditions were used because they represent the clearest contrast in resting-state EEG. In particular, alpha waves (8–12 Hz) increase when the eyes are closed and decrease when the eyes are open. This produces a strong and consistent difference that can be effectively captured by a classifier, even with a relatively small dataset.

In contrast, the mental_arithmetic condition involves additional cognitive processes such as attention and working memory. This introduces greater variability. Including this condition would make the classification task more complex and harder to interpret. Additionally, the two

runs of each condition were merged because they were recorded under identical experimental conditions and represent the same brain state. This increases the number of training samples per class and improves the robustness of the model.

Q4: Whether your features are normalized and what you did if they were not

The features were normalized using StandardScaler. This is because the three feature groups are on very different numerical scales, band power is in μV^2, Hjorth Mobility and Complexity are small dimensionless numbers, and DCT coefficients can be much larger. If these raw values are fed to a classifier, the features with bigger numbers would dominate the decision just because of their scale, not because they are actually more useful. This is fixed using StandardScaler by converting each feature to have a mean of 0 and a standard deviation of 1. This improves the stability and performance of classifiers such as LDA and SVM.

Part 2: Classification (50 Marks)

We selected our two classifiers based on four criteria from the lecture. First, our dataset has around 50 epochs per class, which is below the 100-per-class guideline where LDA is usually recommended. So we chose LDA as our main classifier and used linear SVM for comparison. Second, even though this is not a real-time system, BCI applications usually need fast predictions. Because of that, simpler models like LDA and linear SVM made more sense than more complex ones. Third, interpretability was important to us. We wanted to make sure the model was learning actual brain activity patterns, not just noise. With LDA, we could look at the weights and saw that occipital alpha power was the main feature, which matches what we expect from neuroscience. Finally, we believe the boundary between eyes-open and eyes-closed is mostly linear. This is because the difference is mainly due to alpha suppression at Oz, which is a well-known and consistent brain response.

For these reasons, we didn't think SVM with RBF kernel or deep learning was necessary for this dataset, and LDA with linear SVM was a suitable choice. Based on our results, the LDA classifier achieved 90% accuracy with perfect precision on the eyes-closed class, confirming that alpha power at central and parietal channels was sufficient to separate the two brain states reliably.

2.1 Setting Up the Classifier and Hyperparameters

For LDA, we used default hyperparameters inside a scikit-learn Pipeline with a StandardScaler before it. No manual tuning was needed because LDA calculates everything directly from the data , it means each class and a shared covariance matrix. We chose LDA because it is commonly used for EEG classification in BCI research, works well when classes are linearly separable, and is simple and fast to train, which suits our small dataset of around 50 epochs per class.

For SVM, we ran two variants: linear and RBF kernel both inside a Pipeline with StandardScaler. The linear SVM used default C with class_weight='balanced' to handle any

minor class imbalance. The RBF SVM used C=1.0 and gamma='scale', which automatically sets gamma based on the number of features and data variance. We used class_weight='balanced' for both variants so neither class dominates the decision boundary. The pipeline was used for the same reason as LDA to prevent data leakage during cross-validation.

2.2 5-Fold Stratified Cross-Validation

For both classifiers, we applied the same 5-fold stratified cross-validation object, which splits the data into 5 folds and trains the model 5 times each time using 4 folds for training and 1 fold for testing. Using the same CV object for both models ensures a fair and direct comparison. Stratification ensures each fold keeps the same class ratio as the full dataset, giving a more reliable performance estimate than a single train/test split. We expected both classifiers to perform well above chance because alpha suppression at central and occipital channels is a strong and consistent effect.

2.3 Mean Accuracy ± Standard Deviation

LDA achieved a mean accuracy of 0.836 ± 0.075, SVM linear achieved 0.875 ± 0.072, and SVM RBF achieved 0.876 ± 0.108 all well above the 50% chance level. SVM linear and SVM RBF produced almost identical mean accuracy, suggesting the decision boundary is largely linear and the RBF kernel added no real benefit. However, SVM RBF had a noticeably higher standard deviation (0.108), meaning its performance was less consistent across folds. LDA had the lowest mean accuracy but trained the fastest and showed stable results, making it a strong practical choice given our small dataset.

2.4 Confusion matrix

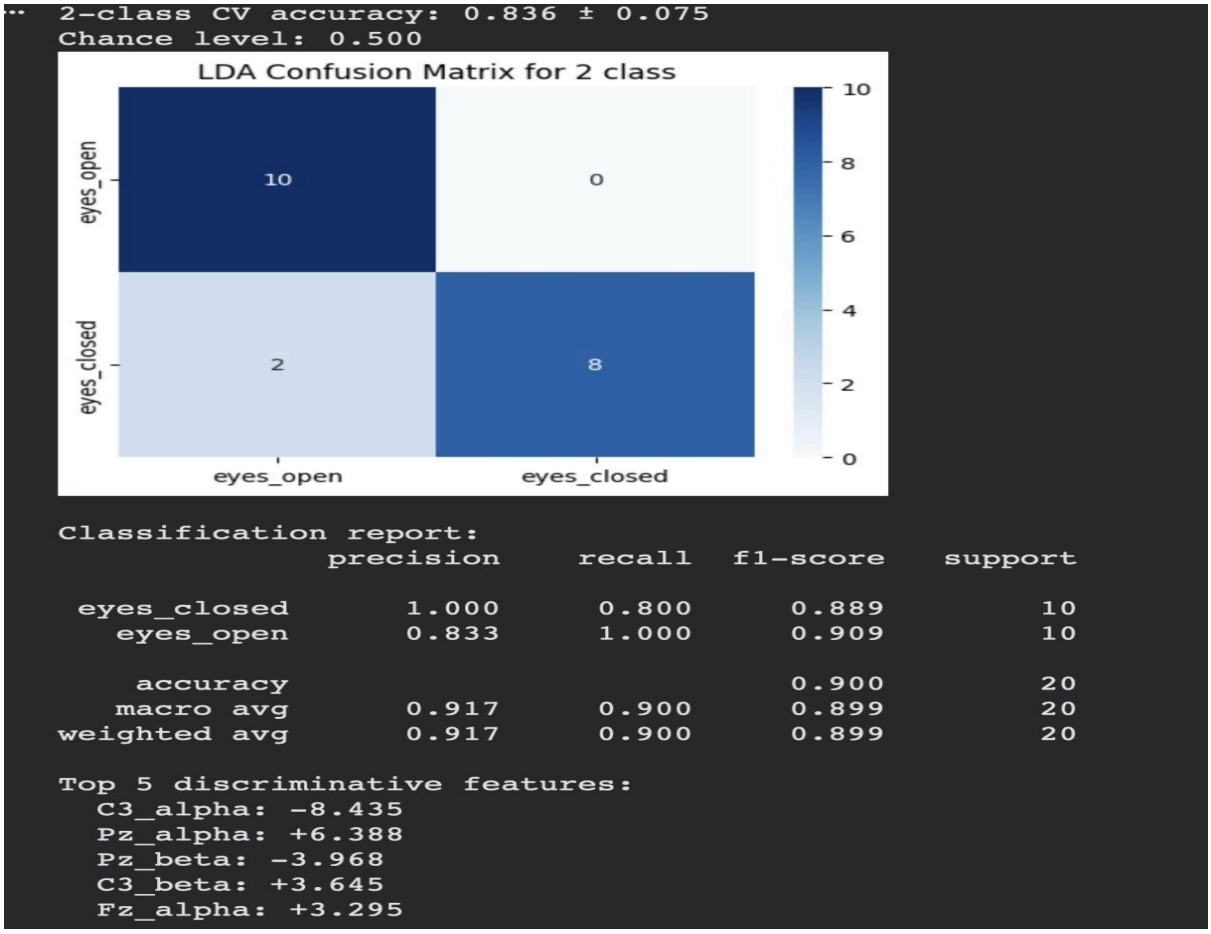


Figure 1: Confusion matrix of LDA for 2 class

The confusion matrix shows that LDA correctly classified all 10 eyes-open epochs and 8 out of 10 eyes-closed epochs. The 2 wrong predictions were eyes-closed epochs that the model thought were eyes-open. The model never confused eyes-open as eyes-closed. This means the model is better at detecting eyes-open than eyes-closed, which makes sense because eyes-open has a very clear alpha suppression signal that is easy to spot.

For eyes-closed, the precision was perfect (1.000), which means every time the model predicted eyes-closed it was correct. But the recall was 0.800, which means it missed 2 actual eyes-closed epochs. For eyes-open, it was the opposite, the recall was perfect (1.000) so it found all eyes-open epochs, but the precision was 0.833 so it sometimes got confused and called an eyes-closed epoch eyes-open instead. The overall test accuracy was 0.900, which is a bit higher than our CV accuracy of 0.836. This is probably because this particular test split was a little easier than average.

2.5 Training time

The computational efficiency of the classification models was evaluated by explicitly measuring their training time. This was implemented by recording the system time immediately before and after the model fitting process using the time module. Specifically, the fit method of each pipeline which comprises preprocessing (standardization) and the classifier itself was executed on the training dataset, and the elapsed time was computed as the difference between the end and start timestamps. This procedure was applied to both the Linear Discriminant Analysis (LDA) model and the Support Vector Machine (SVM) with a radial basis function (RBF) kernel.

The inclusion of training time as a metric provides insight into the computational cost associated with each model, complementing performance measures such as classification accuracy. This is particularly relevant when comparing algorithms, as models with higher predictive performance may incur greater computational overhead. By quantifying training duration, a more comprehensive evaluation of model suitability is achieved, especially in contexts where efficiency and scalability are important considerations.

Part 3: Comparison and Interpretation (30 Marks)

Metric	LDA (Linear Discriminant Analysis)	SVM (RBF Kernel)
2-class CV accuracy (mean)	0.836	0.876
Accuracy std deviation	0.075	0.108
Training time (seconds)	0.0050(seconds)	0.0171(seconds)

Table 1: A summary comparison of two classifiers

Q1: What does your accuracy score tell you about the difficulty of this classification problem? Is a near-chance result a failure of the classifier, a failure of the features, or a property of the data itself?

The achieved accuracy (≈0.84–0.88 in cross-validation and 0.90 on the test set) indicates that this is a relatively easy classification problem, as performance is substantially above the 50% chance level. This states that the features successfully capture strong physiological differences

between the two classes, particularly alpha-band variations. A near-chance result in this context would more likely indicate poor feature representation or noisy data rather than a limitation of the classifier itself. Therefore, the high accuracy confirms that both the data and features are informative for this task.

Q2: Look at your confusion matrices. Which class is more often confused? Explain.

The confusion matrix shows that the eyes_closed class is more often confused, with 2 out of 10 samples misclassified as eyes_open, while all eyes_open samples were correctly classified. This indicates that the model has slightly more difficulty detecting eyes_closed states. A likely explanation is that while eyes_closed is associated with increased alpha activity, this signal may vary across epochs or be weaker in some cases, which makes it less consistently distinguishable than the clearer alpha suppression observed in eyes_open.

Q3: Which classifier performed better on your data? Was the difference meaningful or within the noise of the cross-validation variance? Based on what you learned in Lesson 6 about dataset size and classifier properties, explain why you got this result.

SVM (both linear and RBF) achieved slightly higher mean accuracy (~0.875–0.876) compared to LDA (~0.836), but this difference is small and falls within the cross-validation variance, especially given the higher standard deviation of SVM RBF. Therefore, the improvement is not statistically meaningful.

Based on Lesson 6 we learnt, when the dataset size is small (<100 epochs per class), LDA is recommended because it is a low-variance, stable estimator, while more complex models like SVM RBF could introduce unnecessary variability and risk overfitting. Additionally, Lesson 6 explains that when the decision boundary is approximately linear as is the case for eyes open vs eyes closed due to consistent alpha-band differences which linear models such as LDA and linear SVM tend to perform similarly. This explains why all classifiers achieved comparable performance, with no significant advantage from using a more complex non-linear model.